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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Data Science for Engineers (course)

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Course outline

Week 1 : Assignment 1

The due date for submitting this assignment has passed.

Due on 2025-08-06, 23:59 IST.

Assignment submitted on 2025-08-06, 23:15 IST

1) Which of the following variable names are INVALID in R? ***1 point***

- [ ] 1_variable
- [x] variable_1
- [x] _variable
- [ ] variable@

No, the answer is incorrect.
Score: 0

Accepted Answers:
1_variable
_variable
variable@

2) The function ls() in R will ***1 point***

- ( ) set a new working directory path
- (•) list all objects in our working environment
- ( ) display the path to our working directory
- ( ) None of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:
list all objects in our working environment

Consider the following code snippet. Based on this, answer questions 3 and 4.

```
ID = c(1, 2, 3, 4)

Patient_name = c("Ram", "Shyam", "Nandini", "Maya")

num.patient = 4

patient_list = list(num.patient, ID, Patient_name)
```

3) Which of the following command is used to access the value “Shyam”? ***1 point***

- print(patient_list[3][2])
- print(patient_list[[3]][1])
- print(patient_list[[3]][2])
- print(patient_list[[2]][2])

Yes, the answer is correct.
Score: 1

Accepted Answers:
print(patient_list[[3]][2])

4) What does the following R code produce? ***1 point***

```
x <- c("apple", "banana", "cherry")
x[2]
```

- “apple”
- “banana”
- “cherry”
- Error

Yes, the answer is correct.
Score: 1

Accepted Answers:
“banana”

5) What is the output of following code? ***1 point***

```
x <- 10 + 5 %% 3

typeof(x)
```

- double
- integer
- list
- None of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:
double

unit=21&lesson=32)

Common doubts asked on R Language (Week-1) (unit?unit=21&lesson=33)

Week 1 Feedback Form : Data Science for Engineers!! (unit?unit=21&lesson=153)

Quiz: Week 1 : Assignment 1 (assessment?name=236)

Week 2 ()

Week 3 ()

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6) State whether the given statement is True or False. ***1 point***

The library reshape2 is based around two key functions named **melt** and **cast**.

- True
- False

Yes, the answer is correct.
Score: 1

Accepted Answers:
True

7) What does the following R code return? ***1 point***

```
x <- c(5, 10, 15, 20)
y <- x[x > 10]
y
```

- 5, 10, 15, 20
- 15, 20
- 10, 15, 20
- Error

Yes, the answer is correct.
Score: 1

Accepted Answers:
15, 20

8) What is the output of the following R code? ***1 point***

```
x <- 1
while (x <= 3) {
    print(x)
    x <- x + 1
}
```

- 1, 2, 3
- 0, 1, 2
- 1, 2, 3, 4
- Error

Yes, the answer is correct.
Score: 1

Accepted Answers:
1, 2, 3

Create the data frame using the code given below and answer questions 9 and 10.

student_data =
data.frame(student_id=c(1:4),student_name=c('Ram','Harish','Pradeep','Rajesh'))

9) Choose the correct command to add a column named **student_dept** to the dataframe **student_data**. ***1 point***

- student_datastudent_dept=c(“Commerce”, “Biology”, “English”, “Tamil”)
- student_data[“student_dept”]= c(“Commerce”,“Biology”, “English”,“Tamil”)
- student_dept= student_data[c(“Commerce”,“Biology”,“English”,“Tamil”)]
- None of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:
student_data[“student_dept”]= c(“Commerce”,“Biology”, “English”,“Tamil”)

10) Choose the correct command to access the element **Tamil** in the dataframe **student_data**. ***1 point***

- student_data[[4]]
- student_data[[4]][3]
- student_data[[3]][4]
- None of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:
student_data[[3]][4]

11) The command to check if a value is of numeric data type is ______. ***1 point***

- typeof()
- is.numeric()
- as.numeric()
- None of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:
is.numeric()

12) What will the following R code return? ***1 point***

```
mat <- matrix(1:9, nrow = 3, ncol = 3, byrow = TRUE)
mat[2, 3]
```

- 6
- 5
- 9
- Error

Yes, the answer is correct.
Score: 1

Accepted Answers:
6

13) What is the result of the following R code? ***1 point***

```
mat1 <- matrix(1:6, nrow = 2, ncol = 3)
mat2 <- matrix(7:12, nrow = 2, ncol = 3)
result <- mat1 + mat2
result
```

- [1] 8 10 12
  [2] 14 16 18
- [1] [2] [3]
  [1] 8 12 16
  [2] 10 14 18
- [1] 8 9 10 11 12 13
- Error

Yes, the answer is correct.
Score: 1

Accepted Answers:
[1] [2] [3]
[1] 8 12 16
[2] 10 14 18